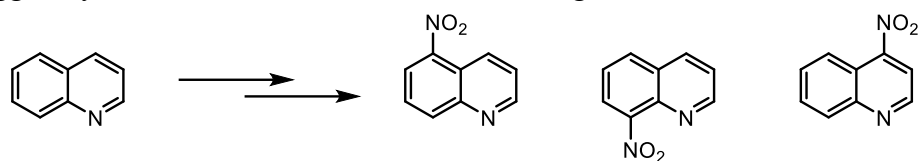
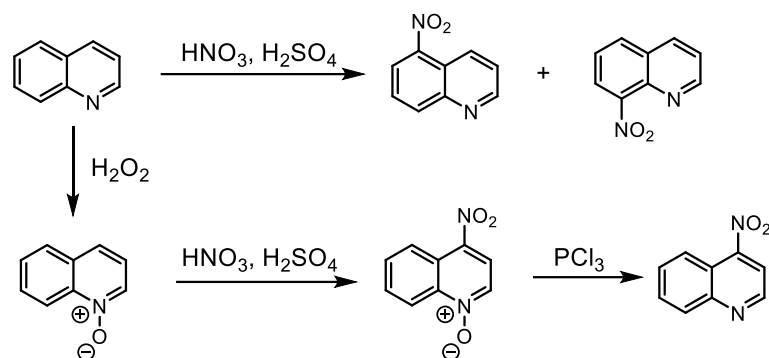


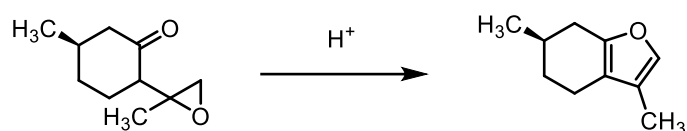
1. Suggest synthetic routes to achieve the following conversions.



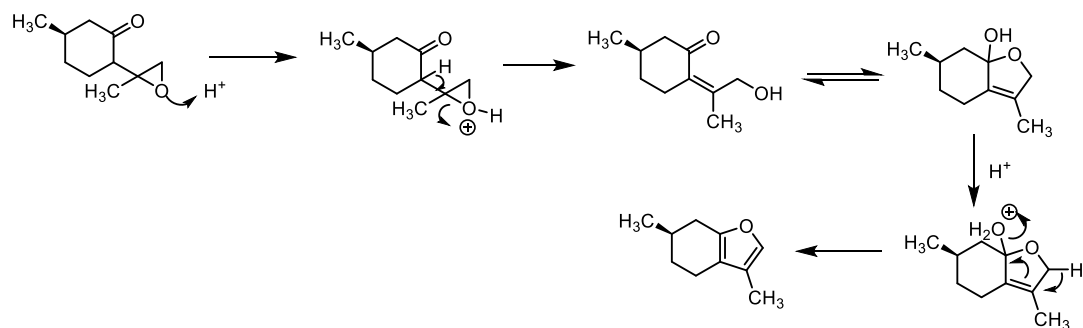
Answer



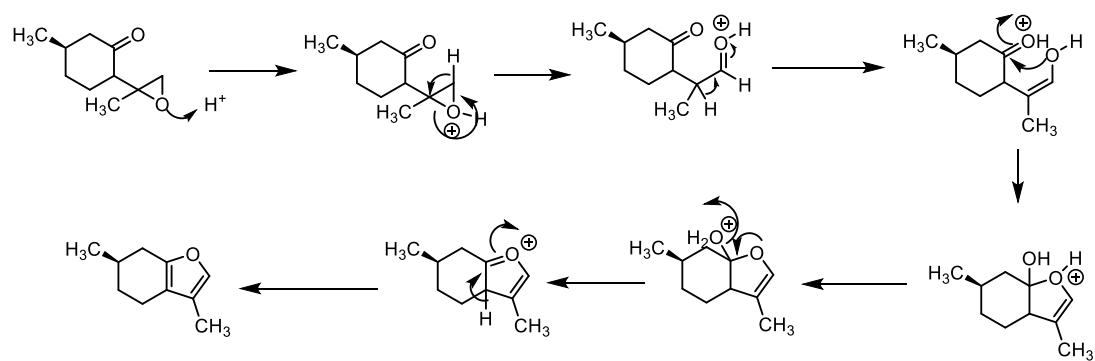
2. Suggest a mechanism for this reaction.



Answer

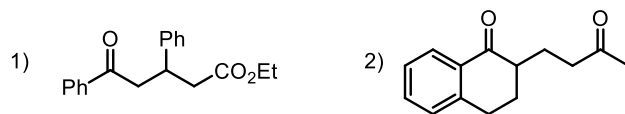


OR

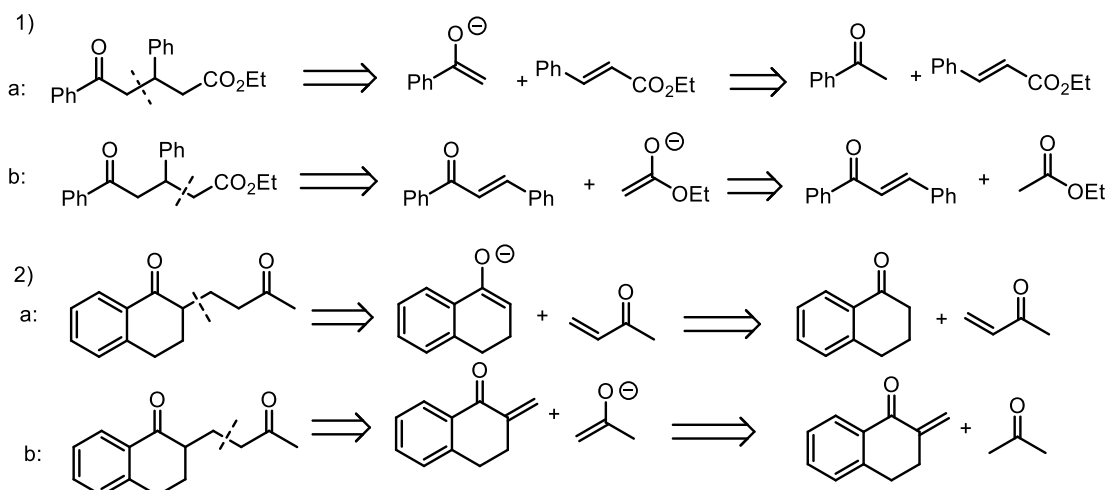


环氧异构化为醛后，从也可从酮烯醇化进攻醛羰基，再脱水。

3. Suggest two different approaches to these compounds by conjugate addition of an enol(ate). Which do you prefer?

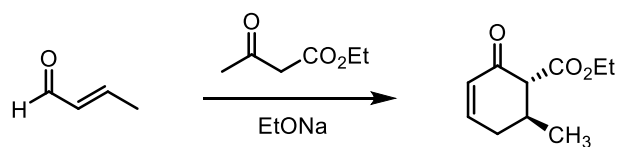


Answer:

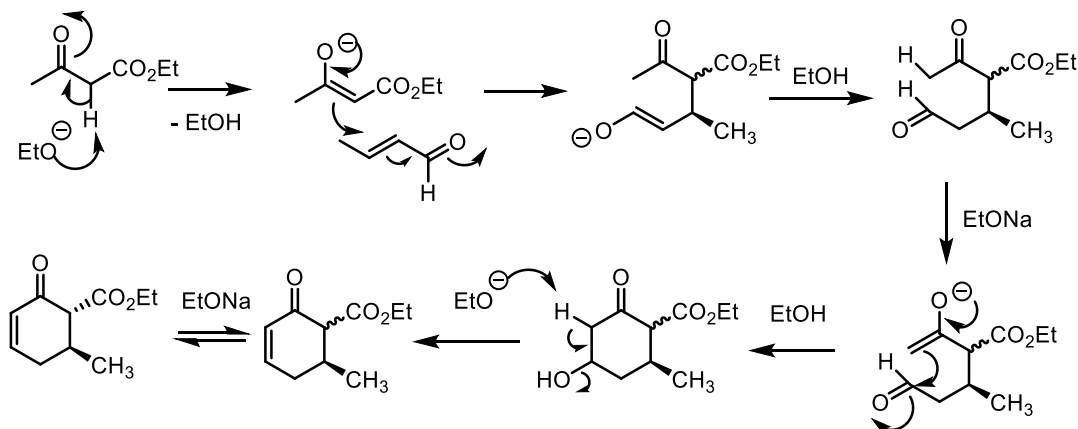


以上都是第一种更好，酮的烯醇负离子更好制备且稳定性更好，反应的共轭加成选择性也会相对好一些。同时也更易制备。

4. Suggest mechanisms for this reaction, commenting on any selectivity.

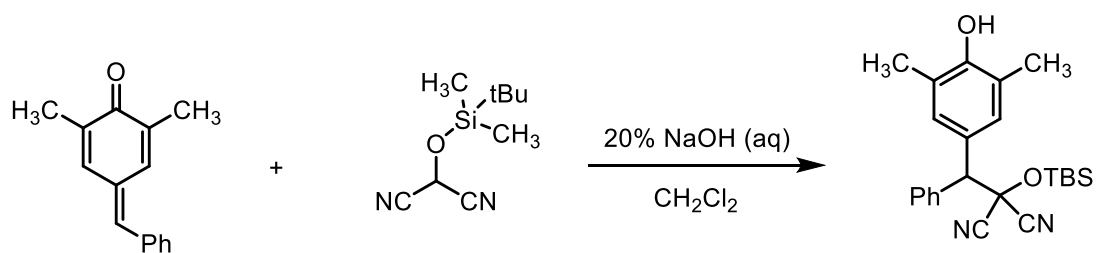


Answer:

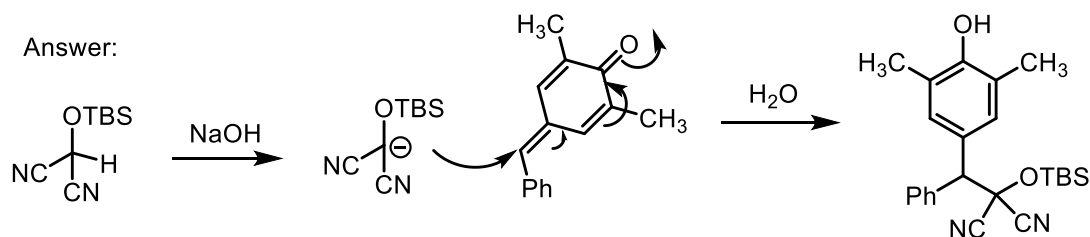


β -羰基酯 pka 最小，去质子化先发生，烯醛可发生 1,2-加成和 1,4 加成，前者可逆，后者不可逆，反应平衡向后者，尤其是生成最终热力学稳定产物的强驱动力。最终产物的两个取代基反式，热力学更稳定。

5. Predict the product for this reaction and suggest mechanisms.

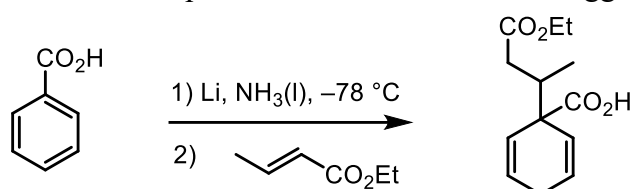


Answer:

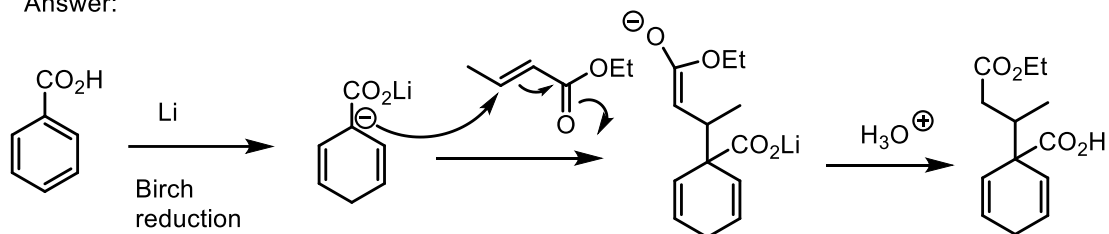


芳香产物热力学驱动位置选择性，发生 1,6-共轭加成。

6. Predict the product for this reaction and suggest mechanisms.

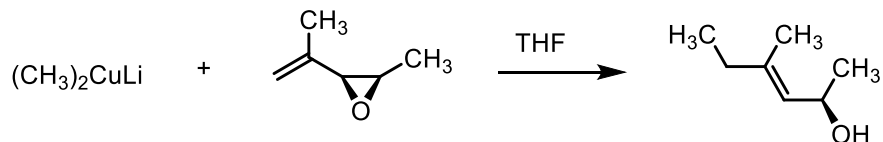


Answer:

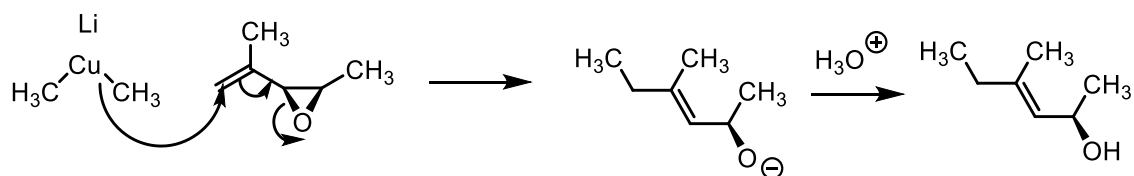


伯奇还原+共轭加成

7. Predict the product for this reaction, commenting on any selectivity.



Answer:



铜锂试剂是软亲核试剂，软亲软，进攻双键，发生 SN²*取代反应

8. Propose a synthetic approach from toluene to this compound using conjugate additions.

